FIITJEE

ALL INDIA TEST SERIES

JEE (Advanced)-2022

CONCEPT RECAPITULATION TEST – III

PAPER –2

TEST DATE: 31-07-2022

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1. ABCD

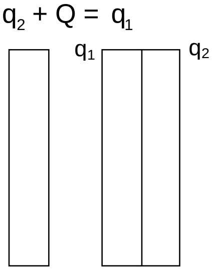
Sol. Before collision, electric field

$$F_C = \frac{(Q-q)Q}{2\varepsilon_0 S}(\text{towards left})$$

During collision, redistribution of charge takes place between B and C, let charges on them are q_1 and q_2

$q_1 + q_2 = Q + q$

$q_2 + Q = q_1$



(Charges on outer most surfaces will be equal)

$$q_1 = Q + \frac{q}{2}$$

$$q_2 = \frac{q}{2}$$

Force on plate C after collision

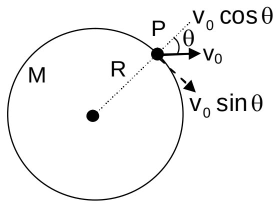
$$= \frac{\left(-Q + Q + \frac{q}{2}\right)}{2\varepsilon_0 S}\frac{q}{2}$$

$= = \dfrac{\left(\dfrac{q}{2}\right)^2}{2\varepsilon_0 S}$ (towards right)

After collision electric field between plates B and C is zero. So kinetic energy of plate C increases and P.E. between the plates remains constant.

2. AD

Sol. From energy conservation



$v_0 = \sqrt{\dfrac{GM}{R}}$

$$-\frac{GMm}{R} + \frac{1}{2}mv_0^2 = -\frac{GMm}{R+h} + \frac{1}{2}mv^2 \quad \text{...(i)}$$

$$mv_o \sin\theta R = mv(R+h) \quad \text{...(ii)}$$

From (i) and (ii) $h = R\cos\theta$

3. ABD

Sol. As radius $r \propto \dfrac{n^2}{z}$

$$\Rightarrow \frac{\Delta r}{r} = \frac{\left(\frac{n+1}{z}\right)^2 - \left(\frac{n}{z}\right)^2}{\left(\frac{n}{z}\right)^2} = \frac{2n+1}{n^2} \approx \frac{2}{n} \propto \frac{1}{n}$$

As energy $E \propto \dfrac{z^2}{n^2}$

$$\Rightarrow \frac{\Delta E}{E} = \frac{\frac{z^2}{n^2} - \frac{z^2}{(n+1)^2}}{\frac{z^2}{(n+1)^2}} = \frac{(n+1)^2 - n^2}{n^2.(n+1)^2}.(n+1)^2$$

$$\Rightarrow \frac{\Delta E}{E} = \frac{2n+1}{n^2} \simeq \frac{2n}{n^2} \propto \frac{1}{n}$$

As angular momentum $L = \dfrac{nh}{2\pi}$

$$\Rightarrow \frac{\Delta L}{L} = \frac{\frac{(n+1)h}{2\pi} - \frac{nh}{2\pi}}{\frac{nh}{2\pi}} = \frac{1}{n} \propto \frac{1}{n}$$

4. BD

Sol. At the bend, when the second train B passes the bend and moves at 60^o to the first train A. The apparent frequency heard by the passenger on train B, is given by $n_a = \left[\frac{V - u_B}{V + u_A \cos 60^o}\right] n$

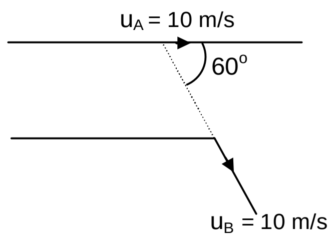
Given V = 300 m/s, $u_A = u_B$ = 36 km/h = 10 m/s.

Thus $n_a = \left[\frac{300 - 10}{300 + 5}\right] 1\,\text{kHz}$

$= \left(\frac{290}{305}\right)(1000) = 950.82$ Hz

The passenger on train A is in the same train as source and so will always hear the source frequency of 1 kHz.

If the train A turns on the bend on the second track while the passenger on train B hears the sound while moving straight on the first track, the apparent frequency heard by him will be



$$n_a = \left[\frac{V - u_B \cos 60^o}{V + u_A}\right] n = \left[\frac{300 - 5}{300 + 10}\right] \text{kHz}$$

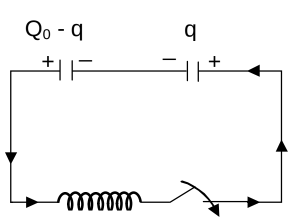
$= \left(\frac{305}{310}\right)$ kHz $= 983.87$ Hz < 1 kHZ

5. AC

Sol. $-L\frac{di}{dt} - \frac{q}{c} + \frac{(Q_0 - q)}{c} = 0$

$$\frac{-Ld^2q}{dt^2} - \frac{2q}{c} + \frac{Q_0}{c} = 0$$

$$\frac{d^2q}{dt^2} = -\frac{2}{LC}\left[q - \frac{Q_0}{2}\right]$$



Frequency = $\frac{1}{\pi\sqrt{2LC}}$

$$= \frac{Q_0}{2}\omega = \frac{Q_0}{2}\sqrt{\frac{2}{LC}} = \frac{Q_0}{\sqrt{2LC}}$$

6. AC

Sol. For first case

Image distance = 3 times object distance

$\therefore \quad \frac{1}{\upsilon} - \frac{1}{u} = \frac{1}{f}$

$\Rightarrow \quad \frac{1}{3\upsilon} - \frac{1}{(-u)} = \frac{1}{f}$

$\Rightarrow \quad \frac{1}{3\upsilon} + \frac{1}{u} = \frac{1}{f}$ …(i)

When distance between screen and lens is increased by 10 cm

$\upsilon = 3u + 10$

$|u'| = \frac{\upsilon}{5}$

$\therefore \quad \frac{1}{\upsilon} - \frac{1}{u} = \frac{1}{f}$

$\Rightarrow \quad \frac{1}{3u+10} - \frac{1}{u'} = \frac{1}{f}$

$\Rightarrow \quad \frac{1}{3u+10} + \frac{5}{3u+10} = \frac{1}{f}$

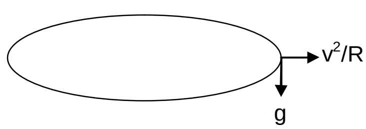
From eqns. (i) and (ii)

$\therefore \quad u = \frac{20}{3}\text{cm}$

$\therefore \quad f = 5$

7. B

Sol.



$$g_{eff} = \sqrt{g^2 + \left(\frac{v^2}{R}\right)} \le 3g$$

$$g^2 + \frac{v^4}{R^2} \le 9g^2$$

$$\frac{100^4}{R^2} \le 8g^2$$

$$R \ge \frac{100^4}{\sqrt{8\times100}} = 100\sqrt{\frac{25}{2}} = \frac{500}{\sqrt{2}}$$

$= 250\sqrt{2}$

8. D

Sol. $\Delta p = \rho g_{eff} h = 10^3 \times 3 \times 10 \times 0.3$

$= 9 \times 10^3$ Pa

9. A

Sol. $f = \frac{n}{2L} \times C$

And $n = \frac{2Lf}{C}$

$n + dn = \frac{2L(f + df)}{C}$

$dn = \frac{2L}{C} df$

10. B

Sol. $P = \int_0^\infty \frac{hf}{e^{\frac{hf}{Tk}} - 1} \times \frac{2L}{C}\ df$

Take $\frac{hf}{kT} = x$

$P = \int_0^\infty \frac{xkT}{e^x - 1} \times \frac{2L}{C} \times \frac{kT}{h} dx = \frac{2k^2T^2L}{Ch} \times \frac{\pi^2}{6}$

$= \frac{\pi^2}{3} \frac{k^2T^2L}{Ch}$

Section – B

11. 4

Sol. From the given information, $\sqrt{v} = m[Z - 1]$

$\Rightarrow \quad \sqrt{v} = 5 \times 10^7 (Z - 1)$

For Z = 41, v = $[5 \times 10^7 \times (40)]^2 = 4 \times 10^{18}$

12. 4

Sol. $2\mu_2 t - \frac{\lambda}{2} = \left(n - \frac{1}{2}\right)\lambda$

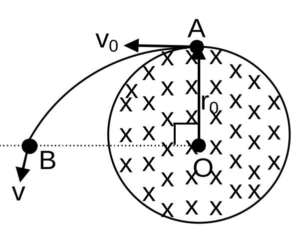
$t = \frac{\lambda}{2\mu_2} = \frac{6400 \times 10^{-10}}{2 \times 1.6}$

$= 2 \times 10^{-7} m = 5K \times 10^{-8}$

K = 4

13. 3

Sol.



$\varepsilon = \int \vec{E}.\vec{dl} = \frac{r_0}{2} \frac{dB}{dt} \times \frac{2\pi r_0}{4} = \frac{\pi r_0^2}{4} \frac{dB}{dt}$

$\therefore$ Work done on charge = $q\varepsilon$

$= \frac{q\pi r_0^2}{4} \frac{dB}{dt} = \Delta KE = \frac{1}{2} m\left(v^2 - v_0^2\right)$

$\frac{2 \times \pi \times 1^2 \times 4}{4} = \frac{1}{2} \times \pi\left(v^2 - 5\right)$

$\Rightarrow \quad v = 3$ m/s

Section – C

14. 00006.67

Sol. Linear impulse momentum theorem

$\int F dt = \Delta p$

$\left[\frac{1}{2}\times 4\times 600+600(\Delta t)\right]\times 2-150\times 10\times \sin 30^{o}\left[4+\Delta t\right]$

$=150\left[0-(-4)\right]$

$\Delta t = 2.67\,\text{sec}$

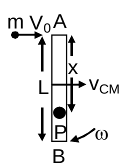
t = $4+\Delta t = 6.67$ sec.

15. 00004.00

Sol. Apply linear impulse momentum theorem as in above case.

16. 00004.00

Sol. Let the linear and angular velocities of centre of mass of rod just after collision be v_{CM} and ω respectively. Applying conservation of linear and angular momentum,



$mv_0 = Mv_{CM}$ ....(i)

$\frac{mv_0 L}{2} = \left(\frac{ML^2}{12}\right)\omega$ ....(ii)

As the collision is elastic, so applying conservation of energy,

$\frac{1}{2}mv_0^2 = \frac{1}{2}Mv_{CM}^2 + \frac{1}{2}\left(\frac{ML^2}{12}\right)\omega^2$ ....(iii)

From Eqn. (i), $v_{CM} = \frac{mv_0}{M}$ and from Eqn. (ii), $\omega = \frac{6mv_0}{ML}$. Substituting in Eqn. (iii) and simplifying,

$\frac{M}{m} = 4$

17. 00001.57

Sol. $\theta = \omega t = \frac{6mv_0}{ML}\times\frac{\pi L}{3v_0}$

$=\frac{6}{3}\left(\frac{m}{M}\right),\pi = \frac{6}{3}\times\frac{1}{4}\pi = \frac{\pi}{2}$

18. 00095.00

Sol. 250 gm $\times 4.18\times(80-75) = (ms)(75-20)$

$(ms) = 95$ J / k

19. 00083.20

Sol. As rate = 500 cal/min. $\therefore$ 5 min = 2500 cal.

$2500\times 4.18 = (750\times 4.18 + 500\times 0.17)\Delta\theta$

$\Rightarrow$ $\Delta\theta = 3.24$

Final temperature = 83.24^{o}C

Chemistry

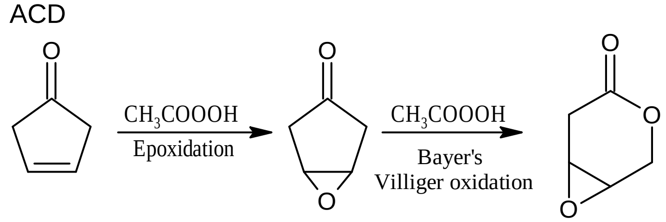
PART – II

Section – A

20. BCD

Sol. Only performic acid [HCOOOH] or HCOOH + H_2O_2 (35%) provides anti hydroxylation.

21. ACD

Sol.


22. AB

Sol. (C) & (D) are not vicinal diols so these alcohols can not show oxidation with IO_4^-.

23. BC

Sol. In (A) 3-bromo-3-methyl pentane is formed while in (D) 3- methyl pentan-3-ol is formed & both are optically inactive as they do not contain chiral corbon atoms.

24. AC

Sol. (A)

$$\frac{760-740}{740} = \frac{1}{n_{H_2O}}$$

$$\Rightarrow n_{H_2O} = 37$$

$$\Rightarrow n_{ice} = 200 - 37 = 163$$

(C)

T_f (original solution = $2 \times \frac{1}{200 \times 18} \times 1000 = \frac{-10}{18}$ °C

25. ABCD

Sol. (A) $d = \frac{Z \times M}{a^3 \times N_A} = \frac{2 \times 120}{64 \times 10^{-24} \times 6 \times 10^{23}}$

$$= \frac{40}{6.4} = \frac{1}{0.16} \text{gm/cm}^3 = 6.25 \text{ gm/cm}^3$$

(B) Number of unit cell = $\frac{\text{Number of particle}}{2}$

$$= \frac{\frac{24}{120} \times 6 \times 10^{23}}{2} = 6 \times 10^{22} \text{ unit cell}$$

(C) $\sqrt{3}a = 4r$

$$\Rightarrow r = \frac{\sqrt{3}}{4} \times 400 = 1.732 \times 100 \text{pm} = 1.732 \mathring{A}$$

(D) $V_{occupied} = 0.68 \times \frac{25}{6.25} = 2.72 \text{ cm}^3$

26. B

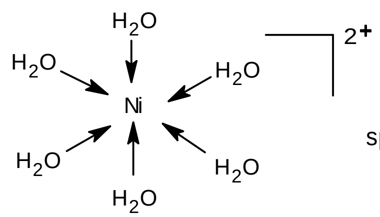
27. A

Sol. (for Q.26-27)

(A)	$CdSO_4$	(B)	CdS
(C)	$BaSO_4$	(D)	$Cd(NO_3)_2$
(E)	$K_2[Cd(CN)_4]$	(F)	$[Cd(NH_3)_4]^{2+}$

28. A

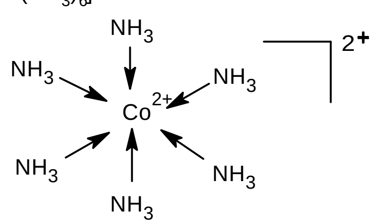
Sol. $[Ni(H_2O)_6]^{2+}$:



sp^3d^2 hybridisation

29. A

Sol. $[Co(NH_3)_6]^{2+}$:



Section – B

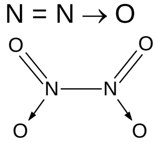
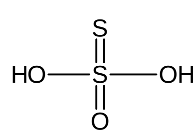
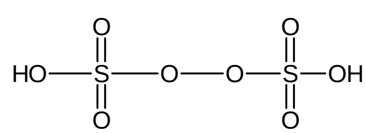
30. 19

Sol. $XeF_4 + O_2F_2 \longrightarrow \underset{Y}{XeF_6} + O_2$

Y has 3 lone pair of electron in each fluorine and one lone pair of electron in xenon. Hence, total lone pair of electrons is 19.

31. 4

Sol. $N = N \rightarrow O$

 ;  ; 

32. 6

Sol. $HA \rightleftharpoons H^+ + A^-$

$\alpha = \frac{1}{100} = 0.01$

$K_a = C\alpha^2$ (by Ostwald's dilution law)

$= 0.01(0.01)^2 = 1\times10^{-6}$

$\therefore$ $pK_a = 6$

100 mL of this 0.01 M acid is treated with 25 mL of 0.01 M $Ca(OH)_2$ or 0.02 M (OH^-),

$Ca(OH)_2 + 2Ha \equiv CaA_2 + 2H_2O$

$[HA] = \dfrac{100 \times 0.01}{1000} = 0.001$ mol

$\left[Ca(OH)_2\right] = \dfrac{25 \times 0.01}{1000} = 2.5 \times 10^{-4}$ mol

HA neutralized by 2.5×10^{-4} mol of $Ca(OH)_2$

$= 5.0 \times 10^{-4}$ mol

Thus, 50% of this acid is neutralized

At this point $[HA] = \left[A^-\right]$

$\therefore \quad pH = pK_a + \log\dfrac{\left[A^-\right]}{[HA]}$

$pH = pK_a = 6$

Section – C

33. 00033.18

Sol. I. $\dfrac{1}{2}N_2(g) + \dfrac{1}{2}O_2(g) \longrightarrow NO, \quad \Delta H = +90.25\,kJ$

II. $NO(g) + \dfrac{1}{2}O_2(g) \longrightarrow NO_2, \quad \Delta H = -57.07\,kJ$

III. $\dfrac{1}{2}N_2(g) + O_2(g) \longrightarrow NO_2, \quad \Delta H = +33.18\,kJ$

$(I + II \equiv III)$

NO_2 is formed in (III) from its elements

Thus, $\Delta_f H^o(NO_2) = +33.18$ kJ mol^{-1}

34. 00114.14

Sol. NO is oxidized to NO_2 in II.

Thus, $\Delta_0 H^o(NO) = -57.07\,kJ\ mol^{-1}$

From Laplace Lavoisier law,

$NO_2 \longrightarrow NO + \dfrac{1}{2}O_2, \ \Delta H^o = +57.07$ kJ

Or $2NO_2 \longrightarrow 2NO + O_2, \ \Delta H = 2 \times 57.07$ kJ

35. 00140.00

Sol. $r = k'[ester];\ k' = 1.8 \times 10^{-3} \times 0.1 \times \dfrac{1000}{18} = 0.01$

$[A]_0 = 1M, [A]_t = 0.25M \Rightarrow t = 2\left(\dfrac{\ln 2}{0.01}\right) = 140$ sec

36. 00091.00

Sol. $\text{Ester} + \underset{(excess)}{H_2O} \xrightarrow{H^+} \text{Acid} + \text{Alcohol}$

$k \times t = \ln\left[\dfrac{V_\infty - V_0}{V_\infty - V_t}\right]$

$0.01 \times 230.3 = \ln\left[\dfrac{100 - 10}{100 - V_t}\right]$

$\Rightarrow \quad V_t = 91$ mL

37. 00400.00

Sol. Bond length = distance between corner and tetrahedral void

$$= \frac{a\sqrt{3}}{4} = 173.2$$

$$a\sqrt{3} = 173.2 \times 4$$

$$\underset{(\text{edgelength})}{a} = \frac{173.2 \times 4}{\sqrt{3}} = 400\,\text{Pm}$$

38. 00002.50

Sol. No. of C-atoms in one unit cell

$$= 6 \times \frac{1}{2} + 8 \times \frac{1}{8} + 4 = 8$$

Density $= \dfrac{8 \times 12}{N_A \times a^3}$ (a = 400 Pm = 400 × 10^{-12} m = 4 × 10^{-8} cm)

$$= \frac{8 \times 12}{6 \times 10^{23} \times \left(4 \times 10^{-8}\right)^3}$$

$$= \frac{960}{6 \times 64} = \frac{160}{64} = 2.5\,g/cm^3$$

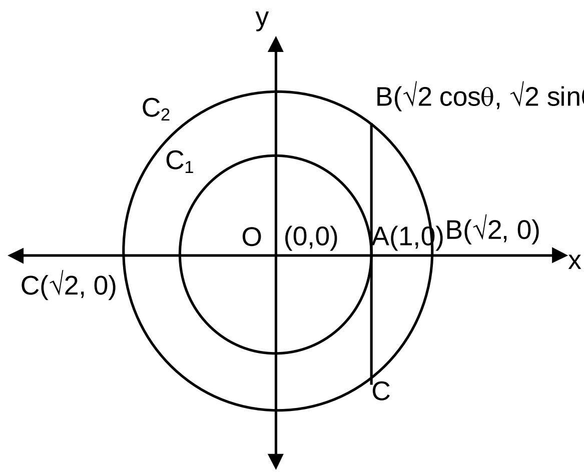
Mathematics

PART – III

Section – A

39. AC

Sol.



We have maximum BC = $2\sqrt{2}$ and minimum BC = 2

$\therefore$ $OA^2 + OB^2 + BC^2 \in [7, 11]$

Let M be the midpoint of AB.

$$\Rightarrow\ M \equiv \left(\frac{1+\sqrt{2}\cos\theta}{2}, \frac{\sqrt{2}\sin\theta}{2}\right) = (h, k)$$

$$\therefore\quad \sin\theta = \frac{2k}{\sqrt{2}}, \cos\theta = \frac{2h-1}{\sqrt{2}}$$

Now on squaring and adding sin θ and cos θ, we get $4k^2 + (2h-1)^2 = 2$

$\therefore$ Locus of M(h, k) is $\left(x - \frac{1}{2}\right)^2 + y^2 = \left(\frac{1}{\sqrt{2}}\right)^2$

40. AB

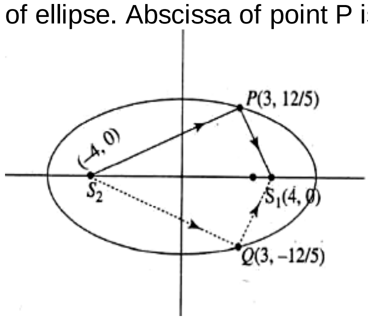
Sol. Given ellipse is $9x^2 + 25y^2 = 225$ or $\frac{x^2}{25} + \frac{y^2}{9} = 1$

$$e = \sqrt{1 - \frac{b^2}{a^2}} = \sqrt{1 - \frac{9}{25}} = \frac{4}{5}$$

Now foci of ellipse are S(± ae, 0) or S(±4, 0).

Given a point (–4, 0) from where a ray is incidenting on ellipse at point P.

Since (–4, 0) is focus of ellipse and therefore, the reflected ray will pass through the second focus of ellipse. Abscissa of point P is 3. Point P lies on the ellipse.



$\therefore\ \frac{3^2}{25}+\frac{y^2}{9}=1$ or $y=\pm\frac{12}{5}$

Now, equation of first reflected ray which passes through two points (4, 0) and $\left(3, \frac{12}{5}\right)$:

$$y-0=\frac{\frac{12}{5}-0}{3-4}(x-4) \text{ or } 5y=-12x+48$$

Now, equation of second reflected ray which passes through the point (4, 0) and $\left(3, -\frac{12}{5}\right)$:

$$y-0=\frac{-\frac{12}{5}-0}{3-4}(x-y) \text{ or } 5y=12x-48$$

41. AD

Sol. $f(xy)=xf(y)+yf(x)-2xy$

Putting x = y = 1, we get f(1) = 2.
Differentiate (1) w.r.t. x keeping y constant, we get

$f'(xy).y=f(y)+yf'(x)-2y$

Putting x = 1, we get

$f'(y).y=f(y)+yf'(1)-2y$

or $yf'(y)=f(y)+y$

or $xf'(x)=f(x)+x$

or $\frac{dy}{dx}-\frac{y}{x}=1$

This is a linear differential equation
I.F. = 1/x

$y\left(\frac{1}{x}\right)=\int\frac{1}{x}dx+c$

$\Rightarrow\ y=x\ln x+cx$

At x = 1, c = 2

$\therefore$ y = x ln x + 2x

42. AC

Sol. In $\Delta ABC, \frac{r}{r_1}=\frac{r_2}{r_3}$

$\frac{s-a}{s}=\frac{s-c}{s-b}$

$s^2-as-bs+ab=s^2-cs$

$ab=(a+b-c)s$

$2ab=(a+b)^2-c^2$

$c^2=a^2+b^2$

$\angle C=90^\circ$

then possible orders of sides are

$c>b>a$	or	$c>a>b$
$s-c<s-b<s-a$	or	$s-c<s-a<s-b$
$r_3>r_2>r_1$	or	$r_3>r_1>r_2$

43. ACD

Sol. (A) $\frac{a}{\sin A} = \frac{b}{\sin B}$

$\Rightarrow$ b is also fixed.

but $\sin C$ may have two different possible values.

if $\sin A = \frac{\sqrt{3}}{2}$

$\Rightarrow$ $A = 60°$ or $120°$

(C) $\frac{a}{\sin A} = 2R$

sinA is fixed but A may have two different values.

(D) $\frac{a}{2R} = \sin A$ (not enough information to form a triangle)

44. AB

Sol. L will be parallel to common line of intersection of planes P_1 and P_2.

45. D

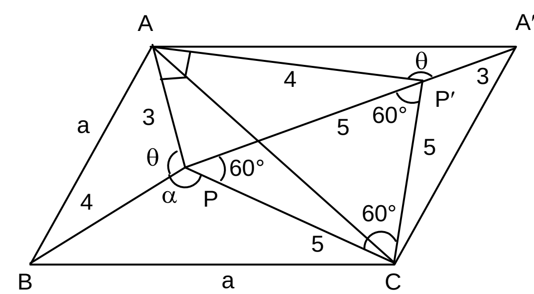
Sol. Clearly AP = A′P′ = 3; BP = AP′ = 4 and PC = P′C = 5

So $\angle CPP' = \angle PP'C = 60°$

$\therefore$ $\Delta PCP'$ is equilateral, PP′ = 5

So $\angle PAP' = 90°$ [$\because$ AP = 3, PP′ = 5]

Area of $\Delta APP'$ is = $\frac{1}{2}(3)(4) = 6$



46. C

Sol. Let $\angle APB = \theta$

In $\Delta APP'$, $\angle APP' = \text{Cos}^{-1}\left(\frac{3}{5}\right) = 53°$

So $\angle AP'P = 37°$

We know $\angle APC = \angle A'P'C = \angle APP' + \angle P'PC = 53° + 60° = 113°$

$\therefore$ $\angle APB = \angle A'P'A = 2\pi - \angle A'P'C - \angle CP'P - \angle PP'A$

$= 2\pi - 113° - 60° - 37°$

$= 2\pi - 210 = 150°$

$\therefore \angle APB = 150°$

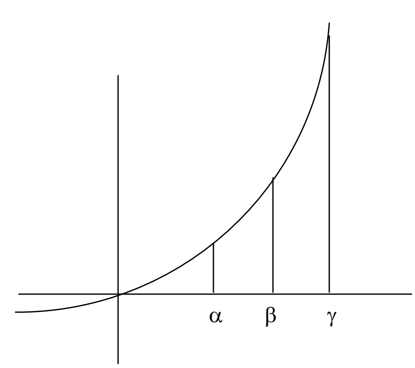
47. B

Sol. Consider g(x) = $\frac{f(x)}{x}$

g′(x) = $\frac{f'(x)x - f(x)}{2} = \frac{f'(x)}{x^2}\left(x - \frac{f(x)}{f'(x)}\right)$

as $\frac{f'(x)}{x^2} > 0$

Let h(x) = $x - \frac{f(x)}{f'(x)}$



$h'(x) = 1 - \frac{f'(x)^2 - f(x)f''(x)}{f'(x)^2}$

$= \frac{f(x)f''(x)}{f'(x)^2} > 0$

$\therefore x - \frac{f(x)}{f'(x)}$ is increasing function.

$x - \frac{f(x)}{f'(x)} > 0 - \frac{f(0)}{f'(0)} = 0$

$\therefore g'(x) > 0.$

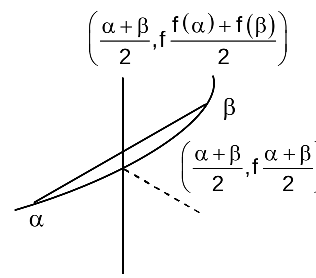
$\therefore \frac{f(\gamma)}{\gamma} > \frac{f(\beta)}{\beta} > \frac{f(\alpha)}{\alpha}$

satisfies A, C, D

48. B

Sol. As the function is concave up

$\frac{f(\alpha)+f(\beta)}{2} > f\left(\frac{\alpha+\beta}{2}\right)$



Section – B

49. 1

Sol. Use L'hospital rule.

50. 4

Sol. $\cos(\cos x - \sin x) = \cos\left(\frac{\pi}{2} - (\sin x + \cos x)\right)$

$\therefore \cos x - \sin x = 2n\pi \pm \left(\frac{\pi}{2} - (\sin x + \cos x)\right)$

Taking + ve sign

$\cos x - \sin x = 2n\pi + \frac{\pi}{2} - \sin x - \cos x$

$\Rightarrow 2\cos x = 2n\pi + \frac{\pi}{2} \Rightarrow \cos x = n\pi + \frac{\pi}{4}$

$n = 0, \cos x = \frac{\pi}{4}$ (only possible values)

$\sin x = \sqrt{1 - \frac{\pi^2}{16}} = \frac{\sqrt{16-\pi^2}}{4}$

Tanking – ve sign

$\cos x - \sin x = 2n\pi - \frac{\pi}{2} - 2n\pi \Rightarrow \sin x = \frac{\pi}{4} - n\pi$

Only possible $\sin x = \frac{\pi}{4}$

As $\frac{\pi}{4} > \frac{\sqrt{16-\pi^2}}{4}$

$\Rightarrow \sin x|_{max,} = \frac{\pi}{4}$

51. 7

Sol. L.C.M(a, b) = 1000 = $2^3.5^3 = 2^{\alpha_1}.5^{\alpha_2}$

L.C.M(b, c) = 2000 = $2^4.5^3 = 2^{\beta_1}.5^{\beta_2}$

L.C.M(c, a) = 2000 = $2^4.5^3 = 2^{\gamma_1}.5^{\gamma_2}$

Max.(α_1, β_1) = 4, Max(β_1, γ_1) = 4, Max(γ_1, α_1) = 4

and Max(α_2, β_2) = 3, Max(β_2, γ_2) = 3, Max(γ_2, α_2) = 3

If $\alpha_1 > \beta_1 \Rightarrow \alpha_1 = 3$

$\beta_1 < 3$ and $\gamma_1 = 4$

$\therefore$ $\alpha_1 = 3$, $\gamma_1 = 4$ and $\beta_1 = 0, 1, 2$ so 3 possibilities

If $\alpha_1 < \beta_1 \Rightarrow \beta_1 = 3$, $\alpha_1 < 3$ and $\gamma_1 = 4$

$\therefore$ $\alpha_1 = 0, 1, 2$, $\beta_1 = 3$, $\gamma_1 = 4$, 3 possibilities

If $\alpha_1 = \beta_1 = 3$ and $\gamma_1 = 4$ $\therefore$ 1 possibilities

$\therefore$ Total 7 possibilities for α_1, β_1, γ_1

If $\alpha_2 > \beta_2 \Rightarrow \alpha_2 = 3$

$\beta_2 < 3$ and $\gamma_2 = 3$ $\therefore$ 3 possibilities

If $\alpha_2 < \beta_2 \Rightarrow \beta_2 = 3$ $\alpha_2 < 3$ and $\gamma_2 = 3 \therefore$ possibilities

If $\alpha_2 = \beta_2 = 3 \Rightarrow \gamma_2 \le 3$ $\therefore$ 4 possibilities

$\therefore$ Total 10 possibilities for α_2, β_2, γ_2

$\therefore$ Total numbers of ordered triplets are $7 \times 10 = 70 = X$

Section – C

52. 00030.00

Sol. a → two entry

b → four entry

c → three entry

Number of c's in diagonal is either 1 or 3

$$= {}^3C_1\left(\frac{\underline{|3}}{\underline{|2}} + \underline{|3}\right) + \frac{\underline{|3}}{\underline{|2}} = 30$$

53. 00006.00

Sol. Number of matrices A = 6

54. 00006.00

55. 00001.44

56. 00576.00

Sol. A – no. of ways in which all are consecutive

B – no. two of them are consecutive.

$$\therefore \quad n(A \cup B) = n(A) + n(B) = [18 - (3-1)] + {}^{(18-3+1)}C_3$$

$$= 16 + 560 = 576$$

57. 00051.00

Sol. Either both are divisible by 3 or both should be selected one from each of other two

$\therefore$ No. of ways = ${}^6C_2 + {}^6C_1 \times {}^6C_1 = 15 + 36 = 51$